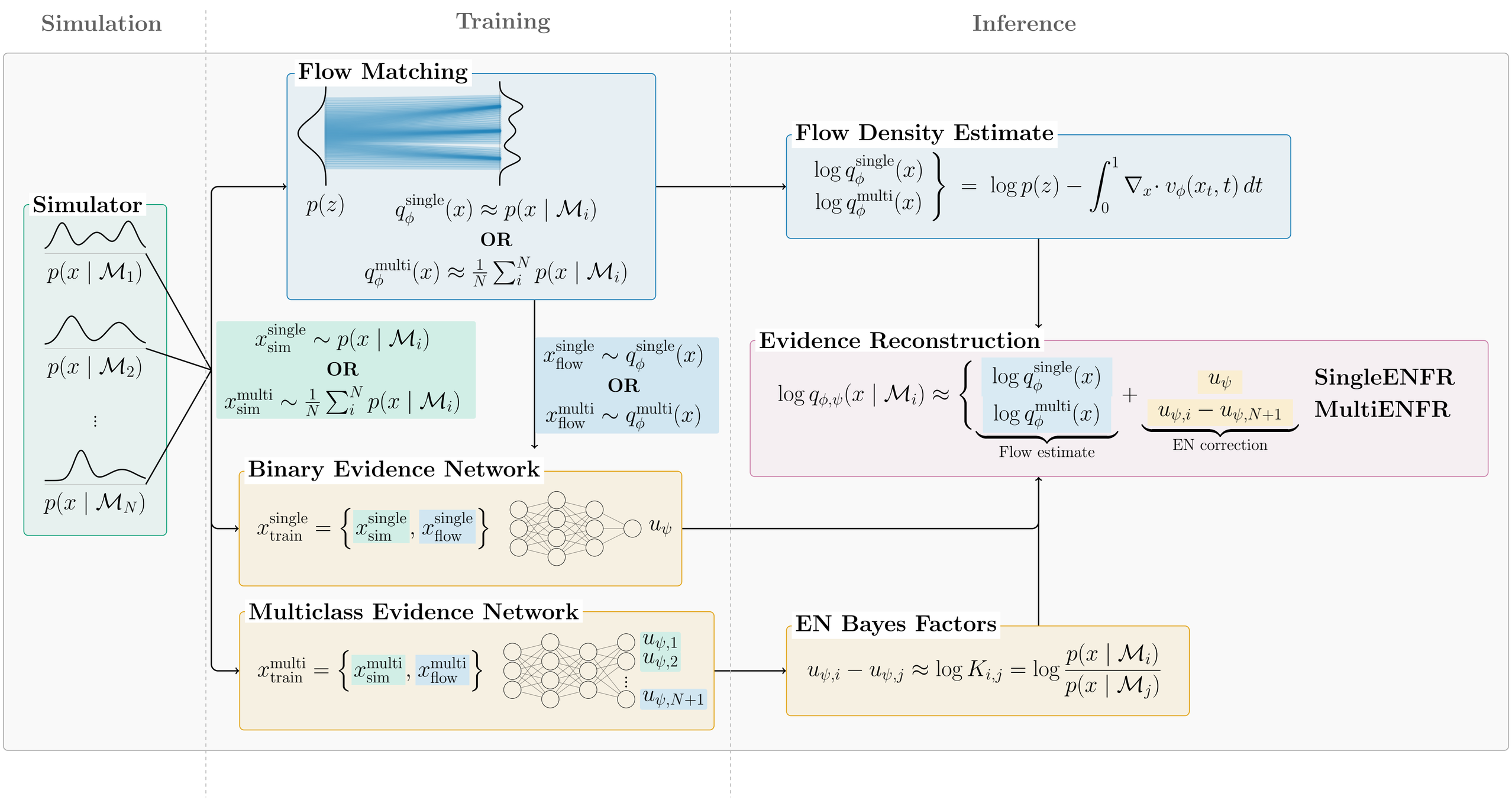

Simulation
Training
Inference
Simulator
$p(x \mid \mathcal{M}_1)$
$p(x \mid \mathcal{M}_2)$
$p(x \mid \mathcal{M}_N)$
Flow Matching
$p(z)$
$q_\phi^{\text{single}}(x) \approx p(x \mid \mathcal{M}_i)$
OR
$q_\phi^{\text{multi}}(x) \approx \frac{1}{N}\sum_i^N p(x \mid \mathcal{M}_i)$
$x_{\text{sim}}^{\text{single}} \sim p(x \mid \mathcal{M}_i)$
OR
$x_{\text{sim}}^{\text{multi}} \sim \frac{1}{N}\sum_i^N p(x \mid \mathcal{M}_i)$
$x_{\text{flow}}^{\text{single}} \sim q_\phi^{\text{single}}(x)$
OR
$x_{\text{flow}}^{\text{multi}} \sim q_\phi^{\text{multi}}(x)$
Binary Evidence Network
$x_{\text{train}}^{\text{single}} = \{x_{\text{sim}}^{\text{single}}, x_{\text{flow}}^{\text{single}}\}$
u_ψ
Multiclass Evidence Network
$x_{\text{train}}^{\text{multi}} = \{x_{\text{sim}}^{\text{multi}}, x_{\text{flow}}^{\text{multi}}\}$
$u_{\psi,1}$
$u_{\psi,2}$
$u_{\psi,N+1}$
Flow Density Estimate
$\log q_\phi^{\text{single}}(x)$
$\log q_\phi^{\text{multi}}(x)$
$= \log p(z) - \int_0^1 \nabla_x \cdot v_\phi(x_t, t)\, dt$
Evidence Reconstruction
$\log q_{\phi,\psi}(x \mid \mathcal{M}_i) \approx$
$\log q_\phi^{\text{single}}(x)$
$\log q_\phi^{\text{multi}}(x)$
Flow estimate
+
u_ψ
$u_{\psi,i} - u_{\psi,N+1}$
EN correction
SingleENFR
MultiENFR
EN Bayes Factors
$u_{\psi,i} - u_{\psi,j} \approx \log K_{i,j} = \log \frac{p(x \mid \mathcal{M}_i)}{p(x \mid \mathcal{M}_j)}$